

FitnessPlanet — Member Retention Strategy

Entity:

FitnessPlanet is a mid-size gym chain that sees a large wave of new members join every January, followed by a steep drop-off in attendance within weeks, internally nicknamed "gym ghosts."

Business Context & Strategic Options: Nearly half of this January's new members had already stopped showing up by February. Leadership is weighing three ways to improve retention before next year's sign-up season:

- **Option A:** Reward every new member for attending group fitness classes, on the assumption that pushing class attendance for everyone will build loyalty.
- **Option B:** Hand out a punch card to every new member that rewards visiting the gym 3 times a week, on the assumption that pushing visit frequency for everyone will build the habit.
- **Option C:** Check whether members paying a higher monthly fee are more likely to leave, to see if adjusting pricing would help retention.

The Task:

- **Phase 1:** Look at how churn changes with class attendance, visit frequency, and distance from the gym to see which of Options A, B, or C the data actually supports.
- **Phase 2:** Assuming management proceeds with Option C, predict churned_by_february from monthly fee and other first-month signals, then isolate how much fee alone drives churn so leadership can judge whether adjusting pricing is worth it.

Dataset Provided: [gym_ghosts_master.csv](#) (50,000 member records across 24 columns)

- **member_id:** Unique member id.
- **age / sex:** Member age (years); sex (F/M/Other).
- **distance_km:** Distance from home to gym (km); nearer tends to be stickier.
- **prior_gym_experience:** 1 if had a gym membership before, else 0.
- **joined_with_friend:** 1 if referred/joined with a friend (accountability).
- **contract_type:** month_to_month / 6_month / annual.
- **monthly_fee_usd:** Monthly membership fee (USD).
- **signup_week_january:** Week of January they signed up (1–4).
- **promo_used:** 1 if joined via the New-Year promo, else 0.
- **primary_goal:** weight_loss / muscle_gain / general_health / event_prep.

- **goal_aggressiveness:** 1 (realistic) to 5 (unrealistic) goal intensity.
- **week1_visits – week4_visits:** Gym visits in each of the first four weeks.
- **avg_weekly_classes:** Average group fitness classes attended per week.
- **booked_induction:** 1 if booked an intro/induction session.
- **personal_trainer:** 1 if using a personal trainer.
- **app_installed:** 1 if installed the gym app.
- **avg_session_minutes:** Average workout session length (minutes).
- **guest_passes_used:** Number of guest passes used.
- **locker_rented:** 1 if rented a locker.
- **churned_by_february:** Target: 1 if ghosted/cancelled by end of February, else 0.

Success Criteria

A clear EDA section demonstrating why two of the options are not as strong, followed by a clear answer on whether pricing is worth adjusting, specific enough for leadership to act on.